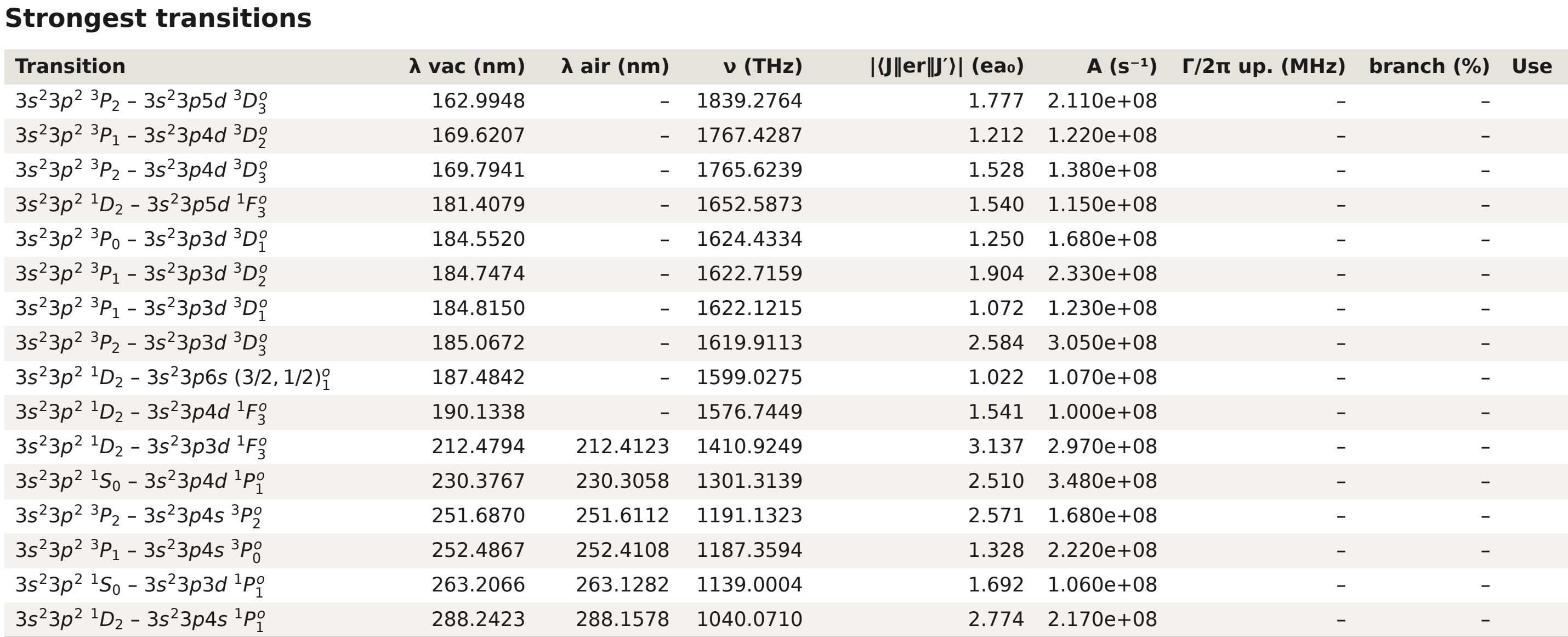
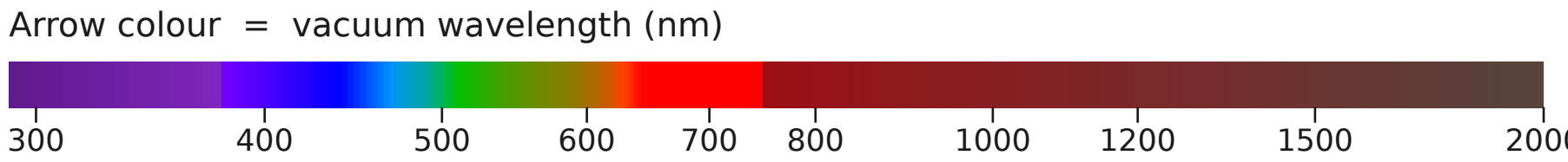


Si (Si I). Strongest classified transitions below 2 μm (NIST): 44 levels, 70 lines.

Si⁺ ionisation limit $65747.76 \text{ cm}^{-1} = 8.15168 \text{ eV} (152.096 \text{ nm})$

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle V | e r | J' J'' \rangle|$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$
 Label caption: NIST energy and measured lifetime (where one exists).



λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.